Non-AI Consequence Gating for IT-OT Integration & Digital Twins in Critical Infrastructure

Architectural Brief on Deterministic Projection ($\Phi \to \Pi_K \to \Omega \to W^2$) across Digital Twin Incident Response, Local LLM/HMI Control, Maritime GNSS Denial, and VSG Byzantine Resilience

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Context: Technical discussion brief for SFI NORCICS at NTNU (Prof. Sokratis Katsikas & Konstantinos Kampourakis), following referral from Dr. Habtamu Abie (Norsk Regnesentral)
Canonical Reference Architecture: `sololys/epistemic-architectures` (Zenodo DOI: `10.5281/zenodo.18436983`)
Public `L1` Software Demonstrators: `sololys/ky-rox-public-demonstrators`
Epistemic Status: `L1` Reference Architecture & Deterministic Software Specification (`INSPIRED_BY != IMPLEMENTS != PROVES`; `PHYSICAL_AUTHORITY = NONE`)

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1. Epistemic Demarcation: Reference Architecture, `L1` Software Verification, and Operational Validation

A central challenge in Cyber-Physical Systems (CPS) and IT-OT integration is the structural asymmetry between **probabilistic IT inference** and **irreversible OT actuation**:

$$\boxed{\text{Generated Candidate } (\mathbf{x_t}) \neq \text{Realized Physical Consequence } \mathbf{x_{t+1}}}$$

In virtual state space, generative models, anomaly detectors, and statistical estimators explore candidate trajectories at near-zero thermodynamic cost ($Q_{\text{math}} = 0.00\,\text{J}$); hypotheses can be revised or rolled back in memory without material damage. Once a command crosses into Operational Technology (OT)—switching relays, slewing maritime rudders, or modulating grid-forming inverters—the action is governed by physical conservation laws and irreversible state erasure ($Q_{\text{phys}} \geq k_B T \ln 2 \cdot \Delta I_{\text{phys}}$).

To keep claims falsifiable and auditable across engineering stages, we enforce a strict three-tier demarcation:

1. **Reference Architecture (`Design & Invariant Specification`):** Formal definition of state spaces, admissible sets $K \subset \mathcal{X}$, projection operators Π_K, and fail-closed verdict lattices.
2. **Software-Level Verification (`L1 Deterministic Demonstrators`):** Reproducible, zero-dependency executable state machines and unit-tested invariant checkers running on deterministic synthetic traces. `L1` software proves that the *code implementation* satisfies the formal specification; it does **not** by itself certify physical hardware or field safety.
3. **Operational Validation (`HIL / Field Qualification`):** Hardware-in-the-loop (HIL), physical interlock timing, galvanic isolation, and certified OT deployment under domain-specific standards.

Between probabilistic IT inference (Φ) and physical OT execution, we interpose a **non-AI, statically verifiable consequence gate**:

$$\Phi(x_t) \xrightarrow{\text{Candidate}} \Pi_K \xrightarrow{\text{Residual } \Delta_K} \Omega \in \{\text{KILL}, \text{HOLD}, \text{OPEN}\} \xrightarrow{\text{Pre-Commit}} W^2 = \langle W_{\text{machine}}, W_{\text{human}} \rangle$$

governed by the strict subsumption order:

$$\boxed{\text{KILL } (0.00\,\text{V DC}) \succ \text{HOLD } (0.00\,\text{V DC}) \succ \text{OPEN } (5.00\,\text{V DC})}$$

where any single invariant violation vetoes actuation unconditionally, and the AI/IT layer holds zero autonomous write authority (`standalone_commit_authority = NONE`, `physical_authority = NONE`, `write_enabled = False`).

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2. Four Industrial IT-OT Domains (SFI NORCICS & CPS4CIP 2026 Alignment)

Domain I — Digital Twins for Incident Detection & Fail-Closed Response (`Door 25` Ontological Twin)

Architectural Bottleneck Conventional Digital Twins in critical infrastructure operate primarily as passive state observers or post-facto statistical anomaly detectors (IDS). When an adversarial command or compromised controller injects a state-corrupting actuation sequence, passive monitoring alerts operators *after* the command has already crossed the OT boundary.

Pre-Execution Involutive Collision & Envelope Gate (Π_K^{Twin}) Rather than treating the Digital Twin as a passive dashboard, **Door 25 (`SPEC-58`)** places the Digital Twin directly inside the pre-commit consequence gate as a **deterministic spectral collision test**:

1. **Involutive Reversibility Check ($S_0 \xrightarrow{a} S_f \xrightarrow{a^{-1}} S_r$):**
   Before any candidate command $a \in \Phi(x_t)$ can reach a physical PLC or relay, the gate clones the last sealed state ($W_{\text{post}} \to W_{\text{twin}}$), simulates the forward transition to S_f, and applies the exact inverse operation a^{-1} to return to S_r. If the round-trip residual exceeds isometric tolerance ($\|S_0 - S_r\| > \varepsilon_{\text{iso}}$), indicating undeclared irreversible state loss or hidden hysteresis, the gate immediately triggers `TWIN_ASYMMETRY_DETECTED` $\implies$ $\Omega \to$ KILL (0.00 V DC).
2. **Admissible Envelope Stress Test (`within_envelope`):**
   During forward simulation inside the Digital Twin, the trajectory is evaluated against hard physical invariants (voltage, thermal, and rate-of-change bounds). Any boundary violation clamps actuation to `TWIN_ENVELOPE_BREACH` $\implies$ $\Omega \to$ HOLD (0.00 V DC) before physical hardware is touched.

Domain II — Locally Deployed LLMs & HMIs in Industrial Control (Session 3)

Architectural Bottleneck When locally deployed Large Language Models (LLMs) or AI-assisted Human-Machine Interfaces (HMIs) translate natural-language operator intent or multi-sensor telemetry into PLC setpoint proposals, hallucinated parameters, unit mismatches, or out-of-envelope transitions can propagate directly into actuator registers if the HMI acts as an unmediated command source.

Non-AI Consequence Gate (Π_K^{HMI}) Every AI/HMI proposal $u_{\text{cand}} \in \Phi(x_t)$ is treated strictly as an untrusted candidate and must pass three deterministic checks before reaching the OT bus:

1. **Forward-Invariant Control Barrier Check (Nagumo Tangency on ∂K):**
   For safety barrier $K = \{x \in \mathcal{X} : h(x) \geq 0\}$, the gate evaluates whether the proposed vector field $f(x_t, u_{\text{cand}})$ points into the tangent cone $T_K(x_t)$:

   $$\langle \nabla h(x_t), f(x_t, u_{\text{cand}}) \rangle + \alpha(h(x_t)) \geq 0.$$

   - If $h(x_t) < 0$ or tangency fails beyond tolerance: $\Omega \to$ KILL (0.00 V DC, `RC-BARRIER-BREACH-KILL`).
   - If $0 \leq h(x_t) \leq \varepsilon_{\text{margin}}$ (boundary transition layer Δ_ε): $\Omega \to$ HOLD (0.00 V DC, `RC-BOUNDARY-MARGIN-HOLD`).
2. **Non-Analytic Transition Latch (Δ_ε):**
   By the Identity Theorem for holomorphic functions, no smooth analytical model can implement an exact binary cutoff (1 on D_{adm}, 0 on D_{elim}) without becoming constant everywhere. The gate therefore enforces an explicit non-analytic transition corridor $[-\varepsilon, +\varepsilon]$ with hardware/software latch hysteresis, preventing high-frequency decision chatter near the boundary.
3. **Dual Pre-Commit Attestation (W^2):**
   Irreversible mode changes require both a deterministic machine invariant receipt (W_{machine}, RFC 8785 canonical JSON + SHA-256 ancestor hash) and an explicit human-at-the-cut-surface authorization token (W_{human}) **prior** to energizing the 5.00 V DC output stage.

Domain III — Maritime GNSS Spoofing & Jamming in Coastal and Port Operations (Session 4)

Architectural Bottleneck In coastal navigation, dynamic positioning (DP), and automated port crane/vessel operations, adversarial GNSS spoofing or RF jamming corrupts external satellite timing and position fixes while

cloud connectivity may be degraded or contested. Systems that equate cryptographic cloud reachability or smooth Kalman-filter innovation with physical truth can be smoothly dragged onto a false trajectory.

Local-First Light-Cone & Phase-Coherence Gate (Π_K^{maritime}) When external GNSS ephemeris is untrusted, physical authority is restricted strictly to the **local causal light-cone** and local physical invariants:

1. **Relativistic Ultra-Wideband (UWB) Time-of-Flight (ToF) Distance Bounding:**
   Peer-to-peer range between vessel and port reference beacons is upper-bounded by the speed of light c in the physical medium:
   $$d_{\text{ToF}} = c \cdot \frac{\Delta t_{\text{round}} - \Delta t_{\text{proc}}}{2} \leq d_{\max}, \qquad |d_{\text{GNSS}} - d_{\text{ToF}}| \leq \tau_{\text{spoof}}.$$
   Because no physical adversary can accelerate electromagnetic propagation faster than c, any spoofed position claim violating the causal ToF light-cone triggers $\Omega \to$ HOLD/KILL (`RC-CAUSAL-LIGHTCONE-VIOLATION`).
2. **Euclidean Phase-Gradient Discrimination ($\mathbb{C}\ \ [-1,1]$):**
   For multi-antenna Angle-of-Arrival (AoA) arrays with baseline d_{ant} and carrier wavelength λ, the measured phase difference $\Delta\phi$ defines the directional sine:
   $$s_{\text{obs}} := \frac{\Delta\phi \cdot \lambda}{2\pi d_{\text{ant}}}.$$
   Any coherent electronic-warfare or spoofing injection that drives $|s_{\text{obs}}| > 1.0$ places $\theta = \arcsin(s_{\text{obs}})$ into the unphysical complex domain $\mathbb{C}\ \ [-1,1]$, immediately triggering `JAMMING_KILL` (0.00 V DC).
3. **Local-First Append-Only State Sealing:**
   All state transitions are chained locally in an append-only SHA-256 ancestor ledger:
   $$W_{t+1} = \mathcal{H}(W_t \parallel \Phi_t \parallel \Delta_K \parallel \Omega_t \parallel \Delta t_{\text{ToF}}),$$
   requiring zero cloud reachability to preserve causal ordering and post-incident forensic auditability.

Domain IV — Virtual Synchronous Generators (VSGs) & Byzantine Fault Tolerance (Sessions 2 & 4)

Architectural Bottleneck Grid-forming Virtual Synchronous Generators (VSGs) emulate rotor inertia and damping in power electronics-dominated microgrids. Sub-threshold Byzantine sensor corruption or coordinated high-frequency oscillatory injection across distributed inverter nodes can excite synthetic swing-equation resonances ($\frac{d^2\delta}{dt^2}$) before traditional static threshold alarms trip.

Relativistic Dispersion &Multiplicative Quorum Gate (Π_K^{VSG})

1. **Dispersion Screening Threshold ($\omega^2 = \mu_{\text{eff}}^2 + c_s^2 k^2$):**
   By subjecting incoming distributed control perturbations $\psi(x,t) \sim e^{i(kx-\omega t)}$ to a screened wave operator with effective mass gap $\mu_{\text{eff}} > 0$:
   $$\left(\frac{\partial^2}{\partial t^2} - c_s^2\nabla^2 + \mu_{\text{eff}}^2\right)\psi = 0 \implies \omega^2 = \mu_{\text{eff}}^2 + c_s^2 k^2,$$
   sub-threshold perturbations with frequency $\omega < \mu_{\text{eff}}$ acquire an imaginary wavenumber $k = i\kappa$ with $\kappa = \frac{\sqrt{\mu_{\text{eff}}^2 - \omega^2}}{c_s} > 0$. Instead of propagating across the inverter network as undamped travelling waves, sub-gap Byzantine chatter becomes **exponentially evanescent** ($e^{-\kappa|x|}$) and is clamped in `HOLD` stasis (0.00 V DC).
2. **Multiplicative 6-Warrant Actuation Barrier:**
   The physical converter modulation command $u(t)$ is gated by a multiplicative Boolean product:
   $$u(t) = u^*(t) \cdot \prod_{m=1}^{6} \mathbf{w}_m = u^*(t) \cdot \left(\mathbf{w}_{\text{risk}} \cdot \mathbf{w}_{\text{quorum}} \cdot \mathbf{w}_{\text{precommit}} \cdot \mathbf{w}_{\text{hw_truth}} \cdot \mathbf{w}_{\text{dispersion}} \cdot \mathbf{w}_{\text{rocof}}\right), \quad \mathbf{w}_m \in \{0,1\},$$
   where $\mathbf{w}_{\text{quorum}} = 1$ requires Byzantine agreement across at least $2f+1$ of $3f+1$ independent sensing channels. If any single warrant drops to 0, the product collapses identically to 0.00 V DC.

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3. Summary Matrix for Joint SFI NORCICS Technical Evaluation

SFI NORCICS / CPS4CIP Domain	Untrusted IT Candidate (Φ)	Non-AI Projection Invariant (Π_K)	Fail-Closed Gate Verdict (Ω)	Verification Tier
Digital Twins (Incident Response)	Compromised controller / OT actuation proposal a	Involutive round-trip $\|S_0 - S_r\| \le \varepsilon_{\text{iso}}$ + envelope stress check in W_{twin}	`TWIN_ENVELOPE_BREACH` (`HOLD`) or `TWIN_ASYMMETRY_DETECTED` (`KILL`, 0.00 V DC)	`L1` Door 25 twin gate (`Python` / `#[no_std] Rust`)
Session 3: Local LLMs & HMIs	Natural-language / AI setpoint proposal u_{cand}	Nagumo barrier tangency $\langle \nabla h, f \rangle + \alpha(h) \ge 0$, Δ_ε spline latch, W^2 receipt	`HOLD` in $[-\varepsilon, +\varepsilon]$; `KILL` (0.00 V DC) if $h(x) < 0$ or unlatched	`L1` deterministic gate suite → PLC/HMI HIL
Session 4: Maritime GNSS Denial	Spoofed/jammed GNSS position & ephemeris	UWB ToF causal light-cone $d_{\text{ToF}} \le c\Delta t/2$, AoA $\|s_{\text{obs}}\| \le 1$, local SHA-256 chain	`JAMMING_KILL` / `HOLD` (0.00 V DC) on light-cone or $\|\sin\theta\| > 1$ breach	`L1` synthetic spoofing traces → coastal/port replay
Sessions 2 & 4: VSG & BFT Grid	Byzantine inverter telemetry & oscillatory chatter	Screened dispersion gap $\omega^2 \ge \mu_{\text{eff}}^2 + c_s^2 k^2 + (2f+1)$ multiplicative warrant	Sub-gap chatter held evanescent in `HOLD`; warrant loss $\Longrightarrow$ KILL (0.00 V DC)	`L1` swing-equation simulator → microgrid testbed

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4. Sammendrag på norsk (Norsk Stamme)

I integrasjonen mellom informasjonsteknologi (IT), Digitale Tvillinger og operasjonell teknologi (OT) i kritisk infrastruktur oppstår en fundamental termodynamisk asymmetri: en sannsynlighetsmodell eller lokal språkmodell (Φ) opererer nesten kostnadsfritt i minnet ($Q_{\text{math}} = 0.00\,\text{J}$), mens et fysisk rorutslag, et reléinnslag eller en omformerordre i strømnettet krysser irreversibilitetsgrensen i romtid ($Q_{\text{phys}} \ge k_B T \ln 2 \cdot \Delta I_{\text{phys}}$).

Ved å plassere en ikke-AI, statisk verifiserbar projeksjons- og konsekvensport ($\Phi \to \Pi_K \to \Omega \to W^2$) mellom IT-inferens og OT-aktuering under subsumpsjonsloven KILL $\succ$ HOLD $\succ$ OPEN, oppnås fire strukturelle garantier på tvers av bruksområdene i SFI NORCICS / CPS4CIP:

1. **Digitale Tvillinger for hendelsesdeteksjon og aktiv respons (`Door 25`):** I stedet for at den digitale tvillingen kun varsler i etterkant, kjøres enhver foreslått styrekommando gjennom en involutiv frem-og-tilbake-kollisjonstest ($S_0 \xrightarrow{a} S_f \xrightarrow{a^{-1}} S_r$) og konvoluttsjekk i tvillingen *før* utgangstrinnet kan strømsettes. Ved avvik låses utgangen til 0.00 V DC.
2. **Lokale LLM-er og HMI-er (Session 3):** Ingen AI-generert kandidat har direkte skrivemyndighet til PLS eller aktuator (`standalone_commit_authority = NONE`). Forslag som bryter Nagumo-tangentkjeglen ved sikkerhetsranden ∂K eller mangler dobbelt forhåndsvitne $W^2 = \langle W_{\text{machine}}, W_{\text{human}} \rangle$, klemmes til 0.00 V DC før de når feltnivået.
3. **Maritim GNSS-manipulasjon (Session 4):** Når satellittsignaler eller skyforbindelse manipuleres eller faller bort i kyst- og havneoperasjoner, begrenses fysisk myndighet til den lokale kausale lyskjeglen via relativistisk UWB Time-of-Flight-avstandsmåling ($d_{\text{ToF}} \le c\Delta t/2$), euklidsk fasediskriminering ($|\sin\theta| \le 1.0$) og lokal-først SHA-256-forfedrekjede.
4. **Virtuelle synkrongeneratorer og BFT (Session 2 & 4):** Ved å håndheve en relativistisk dispersjonsterskel ($\omega^2 = \mu_{\text{eff}}^2 + c_s^2 k^2$) sammen med en multiplikativ $2f+1$-kvorumsbarriere ved kontrollporten, blir sub-terskel bysantinsk støy og høyfrekvent flimmer eksponentielt evanescente og holdes tilbake i 0.00 V DC stase før de kan forplante seg til frekvens- eller RoCoF-ustabilitet i kraftnettet.